```python
            s5_parts.append(pfx + pf_lx + r'\,\Bigl[' + term_lx + r'\Bigr]')
            term_idx += 1
        display(Math(r'\begin{aligned} '
                     + (r' \\[6pt] ').join(s5_parts)
                     + r' \end{aligned}'))

        print()  # visual separator
else:
    print('k < 2: no decomposition to display.')
```

$\boxed{\textbf{Diagram 0} \quad \text{prefactor} = nstar_1, \quad \text{3 edges, 2 with nonzero } \delta}$

Step 1 — full integral:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \; G^R_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_1)\, G^R_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, G^R_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)$$

Step 2 — expand $G^R = c_\delta\,\delta + \Theta\, G^{\rm sm}$:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \; (\delta(-s_3+t_1)+\Theta(-s_3+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_1))\, \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, (\delta(-s_3+t_3)+\Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3))$$

Step 3 — distribute (4 terms):

$$\begin{aligned}
&= nstar_1 \Bigl[\int_{-\infty}^{\infty} ds_3 \; \Theta(-s_3+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_1)\, \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\Bigr] \\[6pt]
&\quad + \; nstar_1 \Bigl[\int_{-\infty}^{\infty} ds_3 \; \delta(-s_3+t_1)\, \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\Bigr] \\[6pt]
&\quad + \; nstar_1 \Bigl[\int_{-\infty}^{\infty} ds_3 \; \delta(-s_3+t_3)\, \Theta(-s_3+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_1)\, \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\Bigr] \\[6pt]
&\quad + \; nstar_1 \Bigl[\int_{-\infty}^{\infty} ds_3 \; \delta(-s_3+t_1)\, \delta(-s_3+t_3)\, \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\Bigr]
\end{aligned}$$

Step 4 — integrate out δ (sifting property):

$$\begin{aligned}
&= nstar_1 \Bigl[\int_{-\infty}^{\infty} ds_3 \; \Theta(-s_3+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_1)\, \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\Bigr] \\[6pt]
&\quad + \; nstar_1 \Bigl[\Theta(-t_1+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1+t_2)\, \Theta(-t_1+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1+t_3)\Bigr] \\[6pt]
&\quad + \; nstar_1 \Bigl[\Theta(t_1-t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(t_1-t_3)\, \Theta(t_2-t_3)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(t_2-t_3)\Bigr] \\[6pt]
&\quad + \; nstar_1 \Bigl[\delta(-t_1+t_3)\, \Theta(-t_1+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1+t_2)\Bigr] \quad \text{(shot noise)}
\end{aligned}$$

Step 5 — stationary variables:

Let $t_1 = 0, \;\; \tau_1 = t_2 - t_1, \;\; \tau_2 = t_3 - t_1, \;\; u = s_3 - t_1$

$$= nstar_1 \Big[\int_{-\infty}^{\infty} du\; \Theta(-t_1 - u)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_1}(-t_1 - u)\, \Theta(-t_1 + \tau_1 - u)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_1}(-t_1 + \tau_1 - u)\, \Theta(-t_1 + \tau_2 - u)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_1}(-t$$

$$+\; nstar_1 \Big[\Theta(\tau_1)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_1}(\tau_1)\, \Theta(\tau_2)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_1}(\tau_2)\Big]$$

$$+\; nstar_1 \Big[\Theta(-\tau_2)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_1}(-\tau_2)\, \Theta(\tau_1 - \tau_2)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_1}(\tau_1 - \tau_2)\Big]$$

$$+\; nstar_1 \Big[\delta(\tau_2)\, \Theta(\tau_1)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_1}(\tau_1)\Big]$$

$\boxed{\textbf{Diagram 1} \quad \text{prefactor} = nstar_2, \quad \text{3 edges, 1 with nonzero } \delta}$

Step 1 — full integral:

$$nstar_2 \int_{-\infty}^{\infty} ds_3\; G^R_{\delta\dot n_1,\tilde n_2}(-s_3 + t_1)\, G^R_{\delta\dot n_2,\tilde n_2}(-s_3 + t_2)\, G^R_{\delta\dot n_1,\tilde n_2}(-s_3 + t_3)$$

Step 2 — expand $G^R = c_\delta\, \delta + \Theta\, G^{\rm sm}$:

$$nstar_2 \int_{-\infty}^{\infty} ds_3\; \Theta(-s_3 + t_1)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_1)\, \big(\delta(-s_3 + t_2) + \Theta(-s_3 + t_2)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_2}(-s_3 + t_2)\big)\, \Theta(-s_3 + t_3)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_3)$$

Step 3 — distribute (2 terms):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3\; \Theta(-s_3 + t_1)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_1)\, \Theta(-s_3 + t_2)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_2}(-s_3 + t_2)\, \Theta(-s_3 + t_3)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_3)\Big]$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3\; \delta(-s_3 + t_2)\, \Theta(-s_3 + t_1)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_1)\, \Theta(-s_3 + t_3)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_3)\Big]$$

Step 4 — integrate out δ (sifting property):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3\; \Theta(-s_3 + t_1)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_1)\, \Theta(-s_3 + t_2)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_2}(-s_3 + t_2)\, \Theta(-s_3 + t_3)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-s_3 + t_3)\Big]$$

$$+\; nstar_2 \Big[\Theta(t_1 - t_2)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(t_1 - t_2)\, \Theta(-t_2 + t_3)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-t_2 + t_3)\Big]$$

Step 5 — stationary variables:

Let $t_1 = 0,\;\; \tau_1 = t_2 - t_1,\;\; \tau_2 = t_3 - t_1,\;\; u = s_3 - t_1$

$$= nstar_2 \Big[\int_{-\infty}^{\infty} du\; \Theta(-t_1 - u)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-t_1 - u)\, \Theta(-t_1 + \tau_1 - u)\, G^{\rm sm}_{\delta\dot n_2,\tilde n_2}(-t_1 + \tau_1 - u)\, \Theta(-t_1 + \tau_2 - u)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-t$$

$$+\; nstar_2 \Big[\Theta(-\tau_1)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-\tau_1)\, \Theta(-\tau_1 + \tau_2)\, G^{\rm sm}_{\delta\dot n_1,\tilde n_2}(-\tau_1 + \tau_2)\Big]$$

$\boxed{\textbf{Diagram 2} \quad \text{prefactor} = nstar_1, \quad \text{4 edges, 2 with nonzero } \delta}$

Step 1 — full integral:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\, G^R_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, G^R_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, G^R_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\, G^R_{\delta v_1,\tilde{n}_1}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\,\delta + \Theta\, G^{\mathrm{sm}}$:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,(\delta(-s_3+t_3) + \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+ t_3))\,(\delta(-s_4+t_1) + \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1))\,\Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(s_3-s_4)$$

Step 3 — distribute (4 terms):

$$= nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\,\Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_3)\,\Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\,\Theta(s_3-s_4)$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_4+t_1)\,\Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\,\Theta(s_3-s_4)$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_3)\,\delta(-s_4+t_1)\,\Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(s_3-s_4)\Big]$$

Step 4 — integrate out δ (sifting property):

$$= nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\,\Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} ds_4\; \Theta(t_2-t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(t_2-t_3)\,\Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\,\Theta(-s_4+t_3)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-s_4+t_3)\Big]$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} ds_3\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\,\Theta(s_3-t_1)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(s_3-t_1)\Big]$$

$$+\; nstar_1 \Big[\Theta(t_2-t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(t_2-t_3)\,\Theta(-t_1+t_3)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-t_1+t_3)\Big]$$

Step 5 — stationary variables:

Let $t_1 = 0,\;\; \tau_1 = t_2 - t_1,\;\; \tau_2 = t_3 - t_1,\;\; u_1 = s_3 - t_1,\;\; u_2 = s_4 - t_1$

$$= nstar_1 \Big[\int_{-\infty}^{\infty} du_1 \int_{-\infty}^{\infty} du_2\; \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1+\tau_1-u_1)\,\Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1+\tau_2-u_1)\,\Theta(-$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} du_2\; \Theta(\tau_1-\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(\tau_1-\tau_2)\,\Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1-u_2)\,\Theta(-t_1+\tau_2-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-t_1+\tau$$

$$+\; nstar_1 \Big[\int_{-\infty}^{\infty} du_1\; \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1+\tau_1-u_1)\,\Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1+\tau_2-u_1)\,\Theta(t_1+u_1$$

$$+\; nstar_1 \Big[\Theta(\tau_1-\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(\tau_1-\tau_2)\,\Theta(\tau_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(\tau_2)\Big]$$

$\boxed{\textbf{Diagram 3}\quad \text{prefactor} = nstar_1,\quad \text{4 edges, 2 with nonzero } \delta}$

Step 1 — full integral:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, G^R_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, G^R_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, G^R_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\, G^R_{\delta v_2,\tilde{n}_1}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\,\delta + \Theta\, G^{\mathrm{sm}}$:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, \left(\delta(-s_3+t_2) + \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\right) \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+ t_3)\left(\delta(-s_4+t_1) + \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\right) \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(s_3-s_4)$$

Step 3 — distribute (4 terms):

$$= nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, \delta(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\, \Theta(s_3-s_4)$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, \delta(-s_4+t_1)\, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(s_3-s_4)$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, \delta(-s_3+t_2)\, \delta(-s_4+t_1)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(s_3-s_4)\Big]$$

Step 4 — integrate out δ (sifting property):

$$= nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} ds_4 \, \Theta(-t_2+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_2+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_4+t_1)\, \Theta(-s_4+t_2)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(-s_4+t_2)\Big]$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(s_3-t_1)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(s_3-t_1)\Big]$$

$$+\ nstar_1 \Big[\Theta(-t_2+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_2+t_3)\, \Theta(-t_1+t_2)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(-t_1+t_2)\Big]$$

Step 5 — stationary variables:

Let $t_1 = 0,\ \ \tau_1 = t_2 - t_1,\ \ \tau_2 = t_3 - t_1,\ \ u_1 = s_3 - t_1,\ \ u_2 = s_4 - t_1$

$$= nstar_1 \Big[\int_{-\infty}^{\infty} du_1 \int_{-\infty}^{\infty} du_2 \, \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-t_1+\tau_1-u_1)\, \Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1+\tau_2-u_1)\, \Theta(-$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} du_2 \, \Theta(-\tau_1+\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-\tau_1+\tau_2)\, \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1-u_2)\, \Theta(-t_1+\tau_1-u_2)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(-t_1$$

$$+\ nstar_1 \Big[\int_{-\infty}^{\infty} du_1 \, \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-t_1+\tau_1-u_1)\, \Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1+\tau_2-u_1)\, \Theta(t_1+u_1)$$

$$+\ nstar_1 \Big[\Theta(-\tau_1+\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-\tau_1+\tau_2)\, \Theta(\tau_1)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_1}(\tau_1)\Big]$$

$\boxed{\textbf{Diagram 4} \quad \text{prefactor} = nstar_2, \quad \text{4 edges, 1 with nonzero } \delta}$

Step 1 — full integral:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, G^R_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\, G^R_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, G^R_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\, G^R_{\delta v_1,\tilde{n}_2}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\,\delta + \Theta\, G^{\rm sm}$:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \; \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,(\delta(-s_3+t_3) + \Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+ t_3))\,\Theta(-s_4+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\,\Theta(s_3-s_4)\, G^{\rm sm}_{\delta v_1,\tilde{n}_2}(s_3-s_4)$$

Step 3 — distribute (2 terms):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \; \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\,\Theta(-s_4+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4$$

$$+ \; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \; \delta(-s_3+t_3)\,\Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_4+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\,\Theta(s_3-s_4)$$

Step 4 — integrate out δ (sifting property):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \; \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-s_3+t_2)\,\Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\,\Theta(-s_4+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4$$

$$+ \; nstar_2 \Big[\int_{-\infty}^{\infty} ds_4 \; \Theta(t_2-t_3)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(t_2-t_3)\,\Theta(-s_4+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\,\Theta(-s_4+t_3)\, G^{\rm sm}_{\delta v_1,\tilde{n}_2}(-s_4+t_3)\Big]$$

Step 5 — stationary variables:

Let $t_1 = 0, \;\; \tau_1 = t_2 - t_1, \;\; \tau_2 = t_3 - t_1, \;\; u_1 = s_3 - t_1, \;\; u_2 = s_4 - t_1$

$$= nstar_2 \Big[\int_{-\infty}^{\infty} du_1 \int_{-\infty}^{\infty} du_2 \; \Theta(-t_1+\tau_1-u_1)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1+\tau_1-u_1)\,\Theta(-t_1+\tau_2-u_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1+\tau_2-u_1)\,\Theta(-$$

$$+ \; nstar_2 \Big[\int_{-\infty}^{\infty} du_2 \; \Theta(\tau_1-\tau_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_1}(\tau_1-\tau_2)\,\Theta(-t_1-u_2)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1-u_2)\,\Theta(-t_1+\tau_2-u_2)\, G^{\rm sm}_{\delta v_1,\tilde{n}_2}(-t_1+\tau$$

$\boxed{\textbf{Diagram 5} \quad \text{prefactor} = nstar_2, \quad \text{4 edges, 1 with nonzero } \delta}$

Step 1 — full integral:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \, G^R_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, G^R_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, G^R_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\, G^R_{\delta v_2,\tilde{n}_2}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\,\delta + \Theta\, G^{\rm sm}$:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4 \; (\delta(-s_3+t_2) + \Theta(-s_3+t_2)\, G^{\rm sm}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2))\,\Theta(-s_3+t_3)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+ t_3)\,\Theta(-s_4+t_1)\, G^{\rm sm}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\,\Theta(s_3-s_4)\, G^{\rm sm}_{\delta v_2,\tilde{n}_2}(s_3-s_4)$$

Step 3 — distribute (2 terms):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4$$

$$+ \ nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \delta(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\, \Theta(s_3-s_4$$

Step 4 — integrate out δ (sifting property):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4$$

$$+ \ nstar_2 \Big[\int_{-\infty}^{\infty} ds_4 \ \Theta(-t_2+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_2+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-s_4+t_1)\, \Theta(-s_4+t_2)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_2}(-s_4+t_2)\Big]$$

Step 5 — stationary variables:

Let $t_1 = 0, \ \ \tau_1 = t_2 - t_1, \ \ \tau_2 = t_3 - t_1, \ \ u_1 = s_3 - t_1, \ \ u_2 = s_4 - t_1$

$$= nstar_2 \Big[\int_{-\infty}^{\infty} du_1 \ \int_{-\infty}^{\infty} du_2 \ \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-t_1+\tau_1-u_1)\, \Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1+\tau_2-u_1)\, \Theta(-$$

$$+ \ nstar_2 \Big[\int_{-\infty}^{\infty} du_2 \ \Theta(-\tau_1+\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-\tau_1+\tau_2)\, \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1-u_2)\, \Theta(-t_1+\tau_1-u_2)\, G^{\mathrm{sm}}_{\delta v_2,\tilde{n}_2}(-t_1$$

Diagram 6 prefactor $= nstar_1$, 4 edges, 2 with nonzero δ

Step 1 — full integral:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ G^{R}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, G^{R}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, G^{R}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, G^{R}_{\delta v_1,\tilde{n}_1}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\, \delta + \Theta\, G^{\mathrm{sm}}$:

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \left(\delta(-s_3+t_2) + \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\right)\left(\delta(-s_3+t_3) + \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\right)\Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(s_3-s_4)$$

Step 3 — distribute (4 terms):

$$= nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \delta(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(s_3-s_4$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \delta(-s_3+t_3)\, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(s_3-s_4$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \delta(-s_3+t_2)\, \delta(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(s_3-s_4)\Big]$$

Step 4 — integrate out δ (sifting property):

$$= nstar_1 \Big[\int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} ds_4 \ \Theta(-t_2+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_2+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(-s_4+t_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-s_4+t_2)\Big]$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} ds_4 \ \Theta(t_2-t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(t_2-t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(-s_4+t_3)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-s_4+t_3)\Big]$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} ds_4 \ \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, \Theta(-s_4+t_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-s_4+t_2)\Big] \quad \text{(shot noise)}$$

Step 5 — stationary variables:

Let $t_1 = 0, \;\; \tau_1 = t_2 - t_1, \;\; \tau_2 = t_3 - t_1, \;\; u_1 = s_3 - t_1, \;\; u_2 = s_4 - t_1$

$$= nstar_1 \Big[\int_{-\infty}^{\infty} du_1 \ \int_{-\infty}^{\infty} du_2 \ \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1+\tau_1-u_1)\, \Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_1+\tau_2-u_1)\, \Theta(-$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} du_2 \ \Theta(-\tau_1+\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-\tau_1+\tau_2)\, \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1-u_2)\, \Theta(-t_1+\tau_1-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-t_1$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} du_2 \ \Theta(\tau_1-\tau_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(\tau_1-\tau_2)\, \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1-u_2)\, \Theta(-t_1+\tau_2-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-t_1+\tau$$

$$+ \ nstar_1 \Big[\int_{-\infty}^{\infty} du_2 \ \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_1}(-t_1-u_2)\, \Theta(-t_1+\tau_1-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_1}(-t_1+\tau_1-u_2)\Big]$$

Diagram 7 prefactor = $nstar_1$, 4 edges, 0 with nonzero δ

All edges purely smooth — no δ decomposition needed.

$$nstar_1 \int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ G^{R}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_2)\, G^{R}_{\delta\dot{n}_1,\tilde{n}_2}(-s_3+t_3)\, G^{R}_{\delta\dot{n}_2,\tilde{n}_1}(-s_4+t_1)\, G^{R}_{\delta v_2,\tilde{n}_1}(s_3-s_4)$$

Diagram 8 prefactor = $nstar_2$, 4 edges, 3 with nonzero δ

Step 1 — full integral:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ G^{R}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, G^{R}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, G^{R}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, G^{R}_{\delta v_1,\tilde{n}_2}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\, \delta + \Theta\, G^{\mathrm{sm}}$:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \ \int_{-\infty}^{\infty} ds_4 \ \left(\delta(-s_3+t_2) + \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\right) \left(\delta(-s_3+t_3) + \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\right) \left(\delta(-s_4+t_1) + \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\right) \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(s_3-s_4)$$

Step 3 — distribute (8 terms):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, \Theta(s_3-s_4)$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_3)\, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, \Theta(s_3-s_4)$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_2)\, \delta(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(s_3-s_4)\Big]$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_4+t_1)\, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(s_3-s_4)$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_2)\, \delta(-s_4+t_1)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(s_3-s_4)\Big]$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_3)\, \delta(-s_4+t_1)\, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(s_3-s_4)\Big]$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_3+t_2)\, \delta(-s_3+t_3)\, \delta(-s_4+t_1)\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(s_3-s_4)\Big]$$

Step 4 — integrate out δ (sifting property):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_4\; \Theta(-t_2+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_2+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, \Theta(-s_4+t_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(-s_4+t_2)\Big]$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_4\; \Theta(t_2-t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(t_2-t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, \Theta(-s_4+t_3)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(-s_4+t_3)\Big]$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_4\; \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_2,\tilde{n}_2}(-s_4+t_1)\, \Theta(-s_4+t_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(-s_4+t_2)\Big] \quad \text{(shot noise)}$$

$$+\; nstar_2 \Big[\int_{-\infty}^{\infty} ds_3\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-s_3+t_3)\, \Theta(s_3-t_1)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(s_3-t_1)\Big]$$

$$+\; nstar_2 \Big[\Theta(-t_2+t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(-t_2+t_3)\, \Theta(-t_1+t_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(-t_1+t_2)\Big]$$

$$+\; nstar_2 \Big[\Theta(t_2-t_3)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_1}(t_2-t_3)\, \Theta(-t_1+t_3)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(-t_1+t_3)\Big]$$

$$+\; nstar_2 \Big[\delta(-t_2+t_3)\, \Theta(-t_1+t_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde{n}_2}(-t_1+t_2)\Big] \quad \text{(shot noise)}$$

Step 5 — stationary variables:

Let $t_1 = 0,\;\; \tau_1 = t_2 - t_1,\;\; \tau_2 = t_3 - t_1,\;\; u_1 = s_3 - t_1,\;\; u_2 = s_4 - t_1$

$$= nstar_2 \Big[\int_{-\infty}^{\infty} du_1 \int_{-\infty}^{\infty} du_2\; \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(-t_1+\tau_1-u_1)\, \Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(-t_1+\tau_2-u_1)\, \Theta(-$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} du_2\; \Theta(-\tau_1+\tau_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(-\tau_1+\tau_2)\, \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot n_2,\tilde n_2}(-t_1-u_2)\, \Theta(-t_1+\tau_1-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde n_2}(-t_1$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} du_2\; \Theta(\tau_1-\tau_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(\tau_1-\tau_2)\, \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot n_2,\tilde n_2}(-t_1-u_2)\, \Theta(-t_1+\tau_2-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde n_2}(-t_1+\tau$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} du_2\; \Theta(-t_1-u_2)\, G^{\mathrm{sm}}_{\delta\dot n_2,\tilde n_2}(-t_1-u_2)\, \Theta(-t_1+\tau_1-u_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde n_2}(-t_1+\tau_1-u_2)\Big]$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} du_1\; \Theta(-t_1+\tau_1-u_1)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(-t_1+\tau_1-u_1)\, \Theta(-t_1+\tau_2-u_1)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(-t_1+\tau_2-u_1)\, \Theta(t_1+u_1)$$

$$+\ nstar_2 \Big[\Theta(-\tau_1+\tau_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(-\tau_1+\tau_2)\, \Theta(\tau_1)\, G^{\mathrm{sm}}_{\delta v_1,\tilde n_2}(\tau_1)\Big]$$

$$+\ nstar_2 \Big[\Theta(\tau_1-\tau_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_1}(\tau_1-\tau_2)\, \Theta(\tau_2)\, G^{\mathrm{sm}}_{\delta v_1,\tilde n_2}(\tau_2)\Big]$$

$$+\ nstar_2 \Big[\delta(-\tau_1+\tau_2)\, \Theta(\tau_1)\, G^{\mathrm{sm}}_{\delta v_1,\tilde n_2}(\tau_1)\Big]$$

Diagram 9 prefactor = $nstar_2$, 4 edges, 1 with nonzero δ

Step 1 — full integral:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; G^{R}_{\delta\dot n_1,\tilde n_2}(-s_3+t_2)\, G^{R}_{\delta\dot n_1,\tilde n_2}(-s_3+t_3)\, G^{R}_{\delta\dot n_2,\tilde n_2}(-s_4+t_1)\, G^{R}_{\delta v_2,\tilde n_2}(s_3-s_4)$$

Step 2 — expand $G^R = c_\delta\,\delta + \Theta\, G^{\mathrm{sm}}$:

$$nstar_2 \int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_3)\, (\delta(-s_4+t_1)+$$
$$\Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot n_2,\tilde n_2}(-s_4+t_1))\, \Theta(s_3-s_4)\, G^{\mathrm{sm}}_{\delta v_2,\tilde n_2}(s_3-s_4)$$

Step 3 — distribute (2 terms):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot n_2,\tilde n_2}(-s_4$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \delta(-s_4+t_1)\, \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_3)\, \Theta(s_3-s_4)$$

Step 4 — integrate out δ (sifting property):

$$= nstar_2 \Big[\int_{-\infty}^{\infty} ds_3 \int_{-\infty}^{\infty} ds_4\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_3)\, \Theta(-s_4+t_1)\, G^{\mathrm{sm}}_{\delta\dot n_2,\tilde n_2}(-s_4$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} ds_3\; \Theta(-s_3+t_2)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_2)\, \Theta(-s_3+t_3)\, G^{\mathrm{sm}}_{\delta\dot n_1,\tilde n_2}(-s_3+t_3)\, \Theta(s_3-t_1)\, G^{\mathrm{sm}}_{\delta v_2,\tilde n_2}(s_3-t_1)\Big]$$

Step 5 — stationary variables:

Let $t_1 = 0,\ \ \tau_1 = t_2 - t_1,\ \ \tau_2 = t_3 - t_1,\ \ u_1 = s_3 - t_1,\ \ u_2 = s_4 - t_1$

$$= nstar_2 \Big[\int_{-\infty}^{\infty} du_1 \int_{-\infty}^{\infty} du_2\, \Theta(-t_1 + \tau_1 - u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1 + \tau_1 - u_1)\, \Theta(-t_1 + \tau_2 - u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1 + \tau_2 - u_1)\, \Theta(-$$

$$+\ nstar_2 \Big[\int_{-\infty}^{\infty} du_1\, \Theta(-t_1 + \tau_1 - u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1 + \tau_1 - u_1)\, \Theta(-t_1 + \tau_2 - u_1)\, G^{\mathrm{sm}}_{\delta\dot{n}_1,\tilde{n}_2}(-t_1 + \tau_2 - u_1)\, \Theta(t_1 + u_1$$

[17]:
```python
#
# 8.1c  Numerical evaluation of Phase J on the   grid
#
# This cell evaluates the Phase J callable (built in cell 8.1) at
# each point on the   grid.  THIS is the slow step - each point
# invokes scipy.integrate.quad on the stripped integrand.
#
# The symbolic decomposition (cell 8.1b) should be inspected first
# to verify the integrand structure looks correct.

if k == 1:
    tau_phase_j = None
    C_tree_phase_j = None
    C_tree_phase_j_slices = None

elif k == 2:
    _total_C_fn = _td_result['total_C']
    tau_phase_j = np.array(tau_residue, dtype=float)
    C_tree_phase_j = np.zeros_like(tau_phase_j, dtype=float)
    for _i, _tv in enumerate(tau_phase_j):
        _val = _total_C_fn(float(_tv), 0.0)
        C_tree_phase_j[_i] = float(np.real(_val))
    # Surviving deltas excluded from smooth evaluation (Ito convention).
    if _td_result['delta_contributions']:
        print(f'  Note: {len(_td_result["delta_contributions"])} surviving '
              f'delta(s) present but excluded from smooth grid.')
    C_tree_phase_j_slices = None
    _mid_j = len(tau_phase_j) // 2
    print(f"Phase J tree  C(0) = {C_tree_phase_j[_mid_j]:.6e}")
    if 'C_tree_residue' in globals() and C_tree_residue is not None and␣
 ↪len(C_tree_residue) == len(tau_phase_j):
        _abs_err = float(np.max(np.abs(C_tree_phase_j - C_tree_residue)))
        print(f"max |Phase J - Phase I residue| = {_abs_err:.3e}")

elif k == 3:
    _total_C_fn = _td_result['total_C']
    tau_phase_j = np.array(tau_residue, dtype=float)
    C_tree_phase_j_slices = {}
```